Skyrim

$W = 4$

weights = $\{1, 3\}$

v gold = $\{5, 2\}$

$n = 2 \rightarrow$ pairings of w, v

table[n][W]

2 4

	n \ W	0	1	2	3	4
$n\{w_n, v_n\}$	0	0	0	0	0	0
$n_1\{1, 5\}$	1	0	5	5	5	5
$n_2\{3, 2\}$	2	0	5	5	5	7

1. Subprob

W = total weight capacity

$n_{(w,v)}$ = pairing where w_n represents weight of gold and v is price of gold

table[n][W]

rows columns

2. Recursive Sol?

base case: if ($w_n > W$) return table[n-1][W]

$n\{w_n, v_n\}$

recursive case: return max (v_n + table[n-1][W - w_n], table[n-1][W]) + recurse call(W, n, table)

$n\{w_n, v_n\}$ $n\{w_n, v_n\}$

Runtime?

The runtime would be $O(nW)$ because you're checking each pair of $\{w_n, v_n\}$ against the total weight capacity W to get a subset of n that gives you the value closest to W.